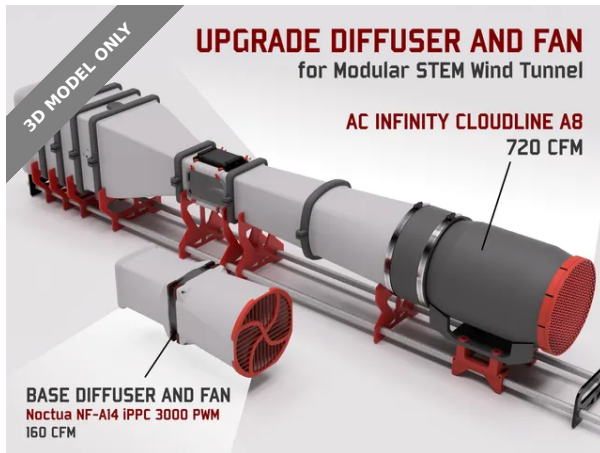




Diffuser and Fan Upgrade Kit for Modular STEM Wind Tunnel



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Summary

Everything required to dramatically increase the flow-rate (and velocity) of the Modular STEM Wind Tunnel.

[Learning](#) > [Engineering](#)

Tags: [diffuser](#) [acinfinity](#) [reprack](#) [cloudline](#) [stemwindtunnel](#) [stemtunnel](#)

Updates

2024-06-18: Added an alternate design for the downstream diffuser section, suitable for a generic fan with a 200mm flow area and face mount. Provided mounting pattern works for [Noctua NF-A20](#) and [EBM-Papst K1G200-AD65-04](#).

2024-05-09: Initial release.

Introduction

This set of designs provides a significant upgrade to the [Modular Wind Tunnel for STEM Education](#), permitting significantly higher flow rate (and

thus, test-section freestream velocity) than the default configuration offers with a 140mm PC fan.

This design utilizes instead an AC Infinity 8-inch AC-powered duct fan [linked here](#) (affiliate link) and below.

Motivation / Benefits

To be clear: this design is **not necessary** for using the Modular Wind Tunnel for its primary use case now -- smoke visualization for basic visual flow diagnostics in a home environment.

Rather, the main motivations for this upgrade are:

1. To enable **future projects** which will offer visual indication or **measurement of forces and moments** acting on test articles.
More velocity = more-significant aerodynamic forces to measure.
2. To improve robustness and flexibility for **permanent installations** that may wish to connect an optional **silencer, filter, and/or flexible exhaust ducting** for management of high-duty-cycle smoke usage in an indoor classroom or museum setting.

Note: this upgrade does **not preclude** the continued use of the wind tunnel for smoke visualization; on the slowest 1 to 3 speed settings (of 10), the fan provides similar performance to the 140 mm PC fan option at full throttle at similar (very quiet) noise levels.

Included Components and Printing Recommendations

Like the largest parts in the default configuration of the Modular Wind Tunnel, these components are designed for printing on a printer with a ~256 x 256 x 256 mm build volume (e.g. Bambu A1, X1, P1 series) or larger.

This design includes:

- An longer two-part diffuser with a continuous square-to-circle transition ending nearly 200 mm in diameter
 - Print one each
 - Can be nested inside each other for printing on one plate
- A new junction ring for mating the two parts
 - Print one ring in two halves
 - Install one **M5x10mm threaded insert per original instructions**
- An optional inline hub diverter and flow straightener
 - Slowly diverts flow away from the fan hub and reduces swirl into the fan plane
 - Prints without supports (sliced in two pieces at the ring), assembled together with pins and/or CA glue

- Installs directly between diffuser and fan, sandwiched into rubber [duct coupling](#).
- **Caution:** carefully check clearances and verify secure attachment.
- A simple fan stand compatible with the AC Infinity Cloudline A8 and the default stands for the wind tunnel
 - Print one stand and 4 countersunk caps
 - Install 4 [M5x10mm threaded inserts](#) for use with M5 flat-head cap screws
- A support cradle set compatible with the RepRack-inspired, EMT conduit-based stand system [offered separately here](#)

Like the largest parts in the default configuration of the Modular Wind Tunnel, these components are designed for printing on a printer with a ~256 x 256 x 256 mm build volume.

Additional Purchase Links

These 8" duct fans from AC Infinity (affiliate links) are compatible with the provided diffuser sections:

- [AC Infinity Cloudline A8](#) - wired 10-speed controller
- [AC Infinity Cloudline S8](#) - slightly higher performance, wired 10-speed controller
- [AC Infinity Cloudline PRO T8](#) - same as S8 but with smart controller

Of these three, the **A8 is recommended as the most cost-effective** and is the model against which the stand feet and diverter provided in this design were created. It includes a basic 10-speed wired controller that is simple to operate with a single button. Join it to the provided diffuser section using a [rubber duct coupler and hose clamps](#).

The S8 and T8 models provide only very slightly higher performance than the A8, and the T8 adds a more-advanced smart controller. This controller, as far as I can tell, provides very little additional benefit to this project unless one wanted to explore the [possibility of Wi-Fi or Bluetooth automation](#).

Model files



Alternate Stand for EMT Conduit (RepRack-inspired)

5 files



emt-conduit-stand-for-upgrade-diffuser-downstream.stl



emt-conduit-stand-for-upgrade-diffuser-upstream.stl



emt-conduit-stand-for-cloudline-a8-fan-and-default-... .stl



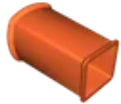
emt-conduit-stand-for-upgrade-diffuser-downstream.dxf

☐ For laser cutting

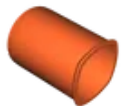


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☐ For laser cutting

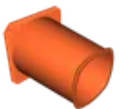


7-upgrade-diffuser-upstream-section.stl



8-upgrade-diffuser-downstream-section-for-8-in-duct... .stl

☐ AC Infinity Cloudline A8 - mate with a rubber duct coupler and hose clamps



8-upgrade-diffuser-downstream-section-for-generic-2... .stl

☐ e.g. Noctua NF-A20 and EBM-Papst K1G200-A65-04



junction-ring-upgrade-diffuser.stl

☐ Print two. Add one M5x10 insert to each.



9-upgrade-fan-guard-ac-infinity-cloudline-a8.stl

☐ Optional endcap. Print inner body with 10-15% hexagonal infill + no walls/upper/lower surfaces



fan-stand-for-ac-infinity-cloudline-a8.stl

☐ Used for both standard stand feet and optional RepRack stand. Qty 1.



fan-stand-cap-washer-ac-infinity-cloudline-a8.stl

☐ Qty 4



hub-diverter-and-stators-for-cloudline-a8.stl

☐ Optional. Deswirl and divert flow away from fan hub. Split inside ring to print w/o supports.

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